Technical Appendix: Key Data Sources and Methods Used

Throughout the book, I draw from three historical data sources.

Most commonly, I use the Stocks, Bonds, Bills, and Inflation data provided by Morningstar and Ibbotson Associates. This data source provides monthly US market returns since 1926 for large-capitalization and small-capitalization stocks, as well as a variety of Treasury bond indexes and a corporate bond index. This is the data source used by William Bengen to formulate the 4 percent rule. Access to this data requires a subscription.

A second data source that I use when discussing stock market valuations is the data collected by Yale professor and Nobel laureate Robert Shiller. On his website, he freely provides monthly updates on US large-capitalization stock returns, dividends, and earnings, as well ten-year Treasury bond yields. This data is available since 1871, making it the longest available data series commonly used for retirement income research. It is also the only data source that is available for free.

Third, when I discuss the international experience of safe withdrawal rates, I rely on Morningstar's Global Returns Dataset. Total returns for stocks, long-term government bonds, and short-term government bonds are available on an annual basis since 1900 for twenty developed-market countries. Morningstar also offers this data set on a subscription basis.

There are three ways that these data sets are generally used to study retirement questions. The simplest method to test a retirement plan is to build a spreadsheet and see how it works assuming a fixed rate of return. In this case, the data may be used to calculate historical average returns for different asset classes, which are then incorporated into the spreadsheet. This approach is also known as deterministic modeling, as there is no randomness in the future outcome. The financial plan could, for example, assume a long-term return on stocks of 10 percent, and that would be the assumed return for each year with no variability over time.

Deterministic approaches are overly simplified because they do not account for volatility (unless the reverse-engineering approach described in this